

AYERTENA BUSINESS AND HEALTH COLLAGE
DEPARTMENT OF ACCOUNTING
BUSINESS STATISTICS –I
WORKSHEET –I

1. Assume Dr. Abebe is an employment of Ayer Tena Health and business collage in the position of research and development department. On January 1, 2019 Dr. Abebe interested to done research in Ayer Tena Health and business collage for the problem of decrease the marks of the student in 2019 academic calendar. Assume the total number of student is 3000 and the researcher selects by using simple random sampling techniques 200 students as a sample from the total population which is 80 accounting, 30 management, 40 marketing and 50 are health department.

Required: Discuss briefly and give short answer of the following question

- a. Define what is the population of the study? What is the population of the above case?
 - b. What is the parameter of the population? List at least three parameter of the population in the above case?
 - c. What is sample of the study? What is the sample of the above study? What is the advantage of sampling?
 - d. What is sampling? List and what is the difference between techniques sampling? What types of sampling techniques are used the above study?
 - e. What is sample size? What is the sample size of the above study?
 - f. What is sampling fream? What is the sampling fream of the above the study?
2. Consider the population of ten at XY students. Last year the total mark for each student was 96, 100, 78, 84, 90, 96, 72, 98 99, and 100. Suppose a sample of four students was selected and their marks were 96, 84, 98 and 100.

Required:

- a. What sampling error in sample survey? What is the reason of sampling error? Calculate the sampling error the above study?
 - b. What is non-sampling error in sample survey? What is the reason of non-sampling error?
3. What is sampling techniques? Discuss seven type of sampling techniques?
4. Population has 200,000 students of whom 50,000 are Arada, 30,000 are Kolfe Keranyo, 10,000 Addis Ketema, and 40,000 are Lideta , Yeka are 30,000 and 40,000 are Bole sub-city. A stratified sampling techniques select sample of size 30,000 should have how many students in Arada, Kolfe, Addis Ketema, Lideta , Yeka or Bole sub-city?
5. Consider a population with size of four ($N = 12$) and it includes variables 2, 7, 9, 10, 12, 8, 16, 14, 20, 3, 5 and 6. If samples with the size of two are to be taken without replacement from the population
- a) Compute the population mean
 - b) How many possible sample arrangements will be there?
 - c) Identify the possible samples.
 - d) Compute the sample means.
 - e) Develop the sampling distribution of the mean.
 - f) Determine the mean of the sampling distribution of the sample mean

6. The standard deviation of total marks for the population of 5000 students is $\sigma = 8000$. Assume that a sample size of the study is 2000; compute the standard deviation of the sample mean?
7. The standard deviation of total marks for the population of 9000 students is $\sigma = 8000$. Assume that a sample size of the study is 900; compute the standard deviation of the sample mean?
8. The standard deviation of the sample mean of the study is 900 students is $x\sigma = 800$. Assume that a sample size of the study is 169; compute the standard deviation of the population mean?
9. Discuss briefly the term statistical inference? What is the advantage of statistical inference? Discuss the life cycle of the sampling prosperities?
10. What is the type of estimator and characteristics of good estimator?
11. The 99% confidence interval estimate of the true population mean marks if a sample of 256 students gives a sample mean of 81. From previous experience we know that the population standard deviation is 900.
Required:
 - a. Estimate and Calculate level of significance of the confidence level
 - b. Estimate and compute Z-value distribution table
 - c. Compute interval estimation of the population mean of the study
12. What is the difference between Z and t distribution? What is the Characteristics of the t and Z distribution table?
13. The 90% confidence interval estimate of the true population mean marks if a sample of 16 students gives a sample mean of 7. The sample standard deviation the study is 9.(Hint $t_{0.05}=2.889$)
Required:
 - a. Estimate and Calculate level of significance of the confidence level
 - b. Calculate degree of freedom
 - b. Estimate and compute t-value distribution table
 - c. Compute interval estimation of the population mean of the study
14. An agricultural chemical retail firm wants to estimate the average number of gallons of weed killer sold per day for the purpose of accurately forecasting and controlling inventory. Twelve business days were monitored, and average daily sales of 10 gallons were recorded. The sample yielded standard deviation of 2 gallons. Calculate the confidence limits at the 95% level?
15. When a sample of 150 students was surveyed regarding the poor marks of the student, 76% believed that decreased marks were due to not preparation and reading of the student.
 - a) Estimate the standard error of the proportion of the student
 - b) Find the upper and lower confidence limits for this proportion, given a 95% confidence level.
16. For a group of 22 college football players, the mean heart rate after a morning workout session was 86 beats per minute, and the standard deviation was 5. Find the 90% confidence interval of the true mean for all college football players after a workout session.

17. The national average for the number of students per teacher for all U.S. public schools is 15.9. A random sample of 12 school districts from a moderately populated area showed that the mean number of students per teacher was 19.2 with a variance of 4.41. Estimate the true mean number of students per teacher with 95% confidence. How does your estimate compare with the national average?
18. A gasoline service station shows a standard deviation of Birr 6.25 for the changes made by the credit card customers. Assume that the station's management would like to estimate the population mean gasoline bill for its credit card customers to be within an error of Birr 1.00. For a 95% confidence level, how large a sample would be necessary?
19. A random sample of shoppers at a convenience store is selected to see how much they spent on that visit. The standard deviation of the population is \$6.43. How large a sample must be selected if the researcher wants to be 99% confident of finding whether the true mean differs from the sample mean by \$1.50?
20. A random sample of 205 college students were asked if they believed that places could be haunted, and 65 responded yes. Estimate the true proportion of college students who believe in the possibility of haunted places with 99% confidence. According to *Time* magazine, 37% of Americans believe that places can be haunted.
21. A survey conducted by Sallie Mae and Gallup of 1404 respondents found that 323 students paid for their education by student loans. Find the 90% confidence of the true proportion of students who paid for their education by student loans.
22. The national average for the percentage of high school graduates taking the SAT is 49%, but the state averages vary from a low of 4% to a high of 92%. A random sample of 300 graduating high school seniors was polled across a particular state area, and it was found that 195 had taken the SAT. Estimate the true proportion of high school graduates in this region who take the SAT with 95% confidence.
23. A researcher wishes to estimate, with 95% confidence, the proportion of people who own a home computer. A previous study shows that 40% of those interviewed had a computer at home. The researcher wishes to be accurate within 2% of the true proportion. Find the minimum sample size necessary.
24. It is believed that 25% of U.S. homes have a direct satellite television receiver. How large a sample is necessary to estimate the true population of homes which do with 95% confidence and within 3 percentage points? How large a sample is necessary if nothing is known about the proportion?